Root Cause Analysis

Problem Statement

BlinkIt has seen a 10% drop in daily active users (DAUs) over the last week. What could be the reasons for this decline, and how would you approach finding the root cause?

Clarification Questions

Understanding the Context:

Q1. When we talk about Daily Active Users (DAUs), does it refer to users who only browse, those who log in, or those who complete a transaction? How do we define "active" in this context?

Answer: DAUs represent users who engage meaningfully with the app on a daily basis, including activities such as browsing products, adding items to their cart, or completing purchases. It goes beyond merely logging in, focusing on interactions that reflect genuine app usage.

Q2. Is the 10% drop in DAUs consistent across all user segments?

Answer: Yes, the drop is consistent across different user cohorts.

Q3. Is this decline uniform across all cities or more prominent in specific regions?

Answer: The decline is more noticeable in metro cities, where a significant portion of our user base resides.

Q4. Have there been changes in user behavior, such as fewer users transitioning from browsing to adding items to their cart?

Answer: Yes, there has been a slight decline in the number of users adding items to their cart after browsing.

Q5. Are users dropping off at specific stages of the user journey, such as during login, product discovery, or checkout?

Answer: We've observed an increase in abandonment rates, particularly during the checkout stage.

Q6. Is this issue present across all platforms (iOS, Android, web), or is it specific to one?

Answer: The issue is observed across all platforms, without any platform-specific anomalies.

Q7. Have other metrics, like session duration or orders, also been affected along with DAUs?

Answer: Yes, session durations have decreased slightly, app uninstall rates have risen, and the number of orders placed has also declined marginally.

Q8. Has a similar drop in DAUs occurred before, potentially indicating a seasonal trend or recurring pattern?

Answer: No similar patterns have been observed in the past.

External Factors

Competition

Q1. Are competitors planning any major promotions in the upcoming weeks?

Answer: Some competitors have launched minor promotions, but they are not substantial enough to explain the recent drop in DAUs.

Q2. Have any competitors or similar services experienced a similar decline in DAUs during this period?

Answer: No, major competitors like Swiggy Instamart and Zepto have not reported similar drops in DAUs.

Q3. Have there been any new competitors entering the market recently?

Answer: No, although the quick-commerce sector remains highly competitive, there haven't been any significant new entrants targeting the same user base.

Political

Q1. Have recent government policies or regulations affected the e-commerce or delivery service industries?

Answer: No, there have been no significant regulatory changes or government actions affecting the food and grocery delivery sector recently.

Q2. Is there any political unrest or regional tension affecting user behavior?

Answer: No, there is no significant political unrest or tension in regions affecting our core user base.

Economic

Q1. Has there been an economic downturn or a shift in consumer spending behavior that might be influencing users?

Answer: While inflation has slightly increased, affecting discretionary spending, essential purchases like grocery shopping have remained stable.

Q2. Have rising logistics or delivery costs impacted service charges, and does this affect DAUs?

Answer: Though fuel prices have risen, leading to a slight increase in delivery charges, internal strategies have mitigated these costs, and they have not had a direct impact on DAUs.

Social

Q1. Are changing social trends, such as minimalist celebrations, influencing BlinkIt users?

Answer: Yes, some users may be opting for more low-key celebrations, reducing demand for certain festive items.

Q2. Are shifting social preferences, like a growing trend toward local produce, impacting BlinkIt's user behavior?

Answer: No significant trends toward local produce have been noted that would directly impact BlinkIt's user base.

Technological

Q1. Are competitors using new technologies that give them a competitive edge over BlinkIt?

Answer: Yes, competitors are leveraging AI for inventory management and route optimization, improving their service quality and potentially attracting BlinkIt's users.

Q2. Have there been any platform or operating system updates that could affect BlinkIt's app functionality?

Answer: No, there have been no recent updates or changes to platforms or operating systems that could impact app functionality.

Environmental

Q1. Are any recent PR or environmental issues affecting BlinkIt's reputation or user perception?

Answer: No, there are no ongoing PR or environmental issues impacting BlinkIt's user perception at the moment.

Q2. Are users increasingly opting for more sustainable delivery options, influencing their choice of service?

Answer: No, there has been no significant shift towards sustainability preferences that are impacting users' choice of delivery service.

Legal

Q1. Have any new legal or compliance changes arisen that could affect BlinkIt's operations or user trust?

Answer: No new legal or compliance issues have been introduced that would impact BlinkIt directly.

Internal Factors

Q1. Are there any inventory shortages affecting product availability?

Answer: Yes, there are inventory limitations, especially for seasonal items like fresh produce and dairy, which may be under pressure due to high demand during festive periods.

Q2. Were there any recent changes to pricing or service fees that might have influenced user sentiment?

Answer: No, there have been no recent changes to pricing or service fees that could have impacted user sentiment.

Q3. Were there any operational disruptions, such as delivery delays or service outages, during the past week?

Answer: There were minor weather-related delays, but nothing significant enough to account for the 10% drop in DAUs.

Q4. Have we received any user feedback or noticed any social media chatter regarding app performance or user experience?

Answer: Yes, there have been user reports on social media about issues with the payment page, where transactions are not completed successfully.

Q5. Has there been an increase in customer support tickets related to payment issues or user frustration over failed transactions?

Answer: Yes, there has been a noticeable uptick in support tickets mentioning payment issues, with users expressing frustration over unsuccessful transactions.

User Journey

1. App Download & Onboarding

- **Action:** Users download the app, allow location access for delivery accuracy, and complete sign-up through OTP verification.
- **Outcome:** A seamless and intuitive onboarding process ensures users are quickly set up and ready to explore the app.

2. Home Screen & Navigation

- **Action:** The home screen provides access to categories, personalized recommendations, and offers, with guided prompts to simplify navigation.
- **Outcome:** Users can effortlessly discover products and features, improving engagement and usability.

3. Search & Product Browsing

- **Action:** Users explore products by browsing categories or using the search bar. They view detailed product pages with images, prices, and nutritional information.
- **Outcome:** A rich and diverse catalog with detailed content makes it easy for users to make informed purchase decisions.

4. Adding Products to Cart

- **Action:** Products are added to the cart with a single tap. Users can easily adjust quantities or remove items directly from the cart.
- **Outcome:** Flexible cart management offers users a sense of control and convenience.

5. Checkout & Payment

- **Action:** Users review their cart, check applicable taxes and delivery fees, and proceed to payment using cards, wallets, or cash on delivery options.
- **Outcome:** Smooth checkout ensures user satisfaction.
- Recent issues have been reported at the payment stage. Many users face failed transactions or unresponsive payment gateways, leading to frustration and abandoned carts.

6. Order Confirmation & Tracking

- **Action:** Upon successful payment, users receive real-time updates on their order status and delivery progress.
- **Outcome:** This transparency ensures users are informed and reassured during the delivery wait time, enhancing trust in the service.

7. Delivery & Post-Purchase

- **Action:** Users receive notifications when their order is out for delivery and can communicate with the delivery personnel for any adjustments. Post-delivery, users rate their experience, redeem offers, and participate in loyalty programs.
- **Outcome:** Feedback mechanisms and rewards improve user satisfaction and encourage long-term retention.

Root Cause Analysis: Payment Issues

Recent reports highlight that many users are experiencing problems during the payment process, specifically at the checkout stage. Upon analysis, the following root causes have been identified:

1. **Technical Failures in Payment Gateway Integration**

- **Description:** Payment gateway systems may encounter issues such as server downtime, transaction timeouts, or slow processing speeds.
- **Impact:** These disruptions lead to failed transactions, leaving users unable to complete their orders.
- **Evidence:** A spike in payment-related errors logged by the system and increased user complaints about failed payments.

2. **Insufficient Real-Time Feedback for Payment Failures**

- **Description:** Users are not receiving clear, actionable feedback when transactions fail, such as the reason for failure or how to resolve it.
- **Impact:** This leads to confusion and reluctance to retry payments, further amplifying cart abandonment rates.
- **Evidence:** A surge in customer support tickets related to unresolved payment issues reflects this gap.

Suggested Improvements for Blinkit

1. Technical Failures in Payment Gateway Integration

Solution:

- **Implement Multiple Payment Gateways:** Integrate multiple payment gateways to serve as backups. If one gateway fails, the system can automatically switch to an alternate, ensuring seamless transactions.
- **Monitor Payment Gateway Performance:** Set up real-time monitoring and alert systems to detect downtime or errors in payment gateways immediately.
- **Optimize Transaction Processing:** Work with gateway providers to reduce transaction timeouts by optimizing API calls and load balancing during peak hours.
- **Conduct Regular Stress Testing:** Periodically test the app's payment processing capabilities under simulated high-traffic conditions to identify and resolve bottlenecks.

Impact:

Increases reliability of transactions and reduces failed payments, improving user trust.

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2. Insufficient Real-Time Feedback for Payment Failures

Solution:

- **Display Detailed Error Messages:** Provide clear, user-friendly error messages that explain the reason for failure (e.g., "Insufficient funds in your card," "Network timeout," or "Payment gateway unavailable").
- **Offer Immediate Recovery Options:** Suggest next steps when a transaction fails, such as retrying the same payment method, choosing an alternative, or opting for Cash on Delivery (COD).
- **Incorporate a Payment Retry Mechanism:** Allow users to retry payments without having to re-enter all details or navigate back through the checkout process.
- **Enhance Communication:** Send real-time updates via SMS or app notifications to confirm payment success or guide users to resolve failed transactions.

Impact:

Reduces user frustration, encourages retrying transactions, and minimizes cart abandonment due to unclear resolution paths.

Metrics to Track Improvement

- **Payment Success Rate:**
  - Tracks the percentage of transactions successfully completed without errors or interruptions, providing insight into the reliability and performance of the payment system.
- **Gateway Downtime Monitoring:**
  - Measures the frequency and duration of payment gateway outages, helping to assess the impact of system reliability on user experience and payment flow.
- **Average Transaction Time:**
  - Monitors the time taken for a payment to be processed from initiation to confirmation, identifying areas where optimization can improve speed and reduce user drop-offs.
- **Gateway Failover Success Rate:**
  - Evaluates the percentage of transactions successfully processed through backup gateways during primary gateway failures, ensuring continuity in the payment process.
- **User Interaction with Error Messages:**
  - Tracks how often users act on error messages by retrying payments or selecting alternative methods, indicating the effectiveness of actionable feedback in reducing cart abandonment.

Summary

By implementing these focused improvements and monitoring the proposed metrics, Blinkit can effectively tackle the DAU decline while enhancing the overall user experience. Prioritizing user feedback, optimizing app functionality, and providing attractive incentives will not only improve user satisfaction but also

foster greater loyalty. These efforts will help reverse the DAU dip, driving higher engagement and reinforcing Blinkit's position in the market.